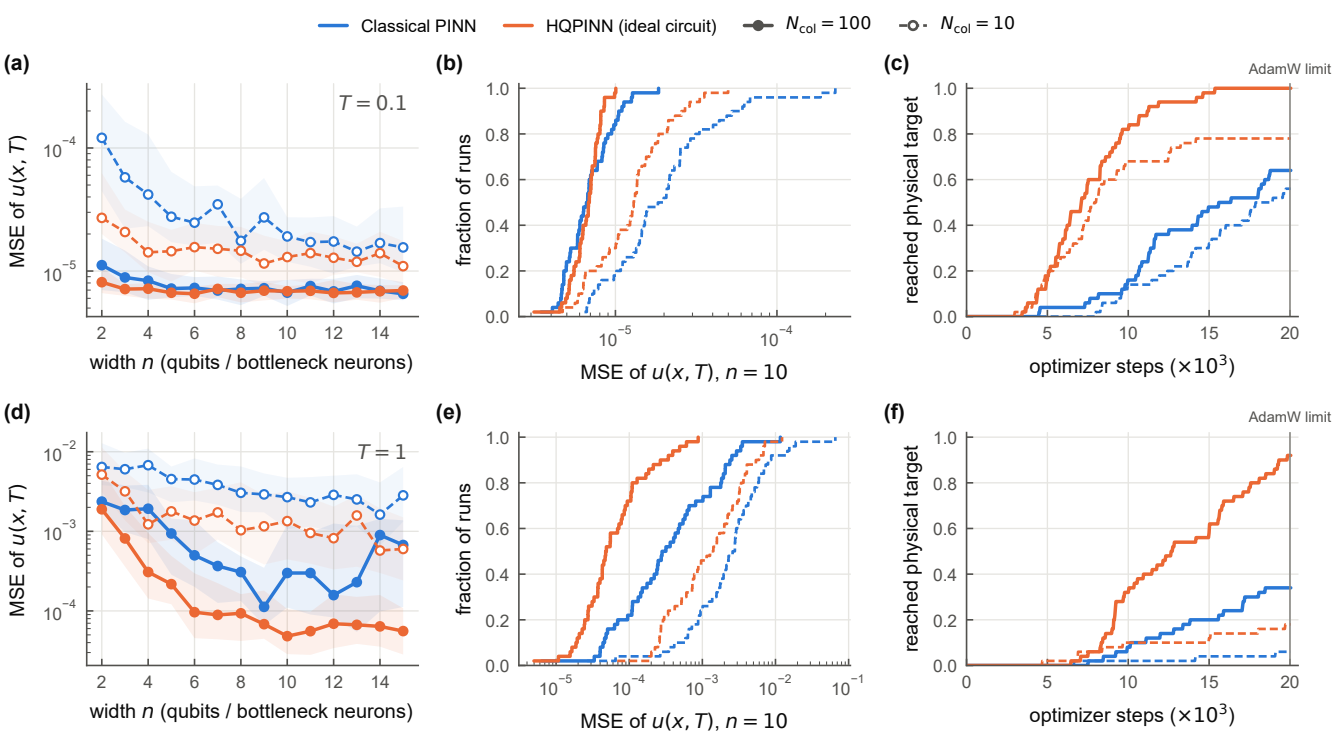

Classical PINN
HQPINN (ideal circuit)
$N_{col} = 100$
$N_{col} = 10$
(a)
$T = 0.1$
MSE of $u(x, T)$
width n (qubits / bottleneck neurons)
(b)
fraction of runs
MSE of $u(x, T)$, $n = 10$
(c)
AdamW limit
reached physical target
optimizer steps ($\times 10^3$)
(d)
$T = 1$
MSE of $u(x, T)$
width n (qubits / bottleneck neurons)
(e)
fraction of runs
MSE of $u(x, T)$, $n = 10$
(f)
AdamW limit
reached physical target
optimizer steps ($\times 10^3$)